



BUCKINGHAMSHIRE
NEW UNIVERSITY

EST. 1891

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Computing essentials
FY007

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Course Work:

App design and development

Introduction

For this assignment, I have been tasked with creating a website or app from conception to a working prototype, detailing, documenting, and displaying all stages of the design process. For this assignment, I have chosen to create an app that I have wanted to make for a long time as I believe it can and will have real-world application in its field. I have spent the majority of my career in the hospitality industry, working my way up from the bottom i.e. washing up and cleaning, to the top levels of management and shareholding. Through the years I realized that there is one particular task that is consensually laborious amongst all positions in the sector, and that task is stock counting and ordering. The idea is to create an app that can link in real-time to the point of sale systems used by various companies to relay information such as best and worst sellers, product par levels, staff upselling, and more, but also to have the ability to detect when a product is low and automatically order said products. The idea is that management would manually adjust this setting, for example, setting Budweiser beer to a par level of 3 cases or 72 bottles, my app detects that Budweiser beer stock currently sits at half a case and automatically adds the par level difference to the next wet stock order. The app will also have many other features depending on the user's position of authority within the company, displaying things like P&L and other financial information. Now I will show my process and design for this idea.

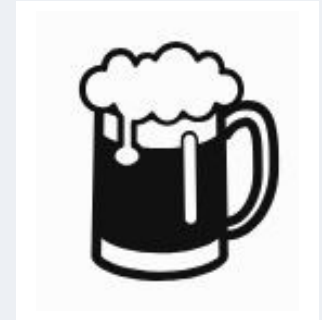
Conception

I have the concept of what I would like my app to be and exactly what it does. My app is designed to save time and decrease the cost of labor, increasing productivity and profits, but also removing the painstaking task of counting stock and manually ordering. The app itself once fully operational will be able to place orders on management's behalf, analyze sell trends, track all individual stock items, and create profit and loss graphs that are clear and easy to understand all while being very user-friendly but also allowing for different levels of access based on a staff member's position. This means that the app will merge with each staff member's point of sale log, but also bar staff can use the app to show them how many per item are needed to bottle up the fridges, while senior management will be able to see which staff member has been the most active in sales, determine what products are selling well, manually adjust automatic ordering suppliers and make well-informed decisions based on the profit and loss displays, and all of this in real-time. I believe this app would be a huge hit in the industry, I would make the app free to download but require a subscription to use as this level of complexity in an app will require regular maintenance and updates. This will also ensure that the app itself generates revenue on a monthly regular basis, allowing for further research and advancements for the app and beyond..

Firstly, the app needs a name and a logo. The term used to fill the fridges is known as 'Bottling up' this is widely used in all hospitality industries. After looking to see if the name had been trademarked, I discovered it is not, so moving forward the app will be called '**Bottle up**' and I will refer to the app as such moving forward.

Next is a logo, doing research into already successful application logos, I noticed that simplicity is key, something bold and striking yet easy to recognise and remember.

I decided to go with black and white as it is the most visually striking based on contrast, but also with simplicity in mind. I believe this logo will be easily recognisable, but also light and fun adding to the user-friendly concept. I have determined that this will be the final design for the logo.



Next is the name of the app but in logo form. I decided to keep the black and white contrast throughout the logo design, again, simplicity is key.

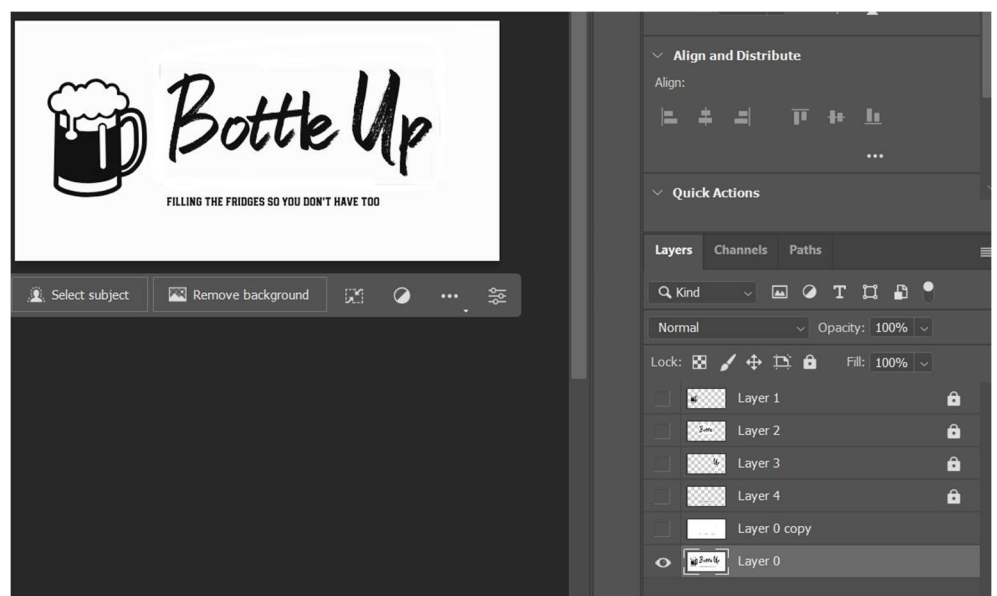
The brush stroke effect and the character design are again meant to be open and fun, showing potential users that the app may be complex but is also user-friendly.

This design ideology is supposed to avoid deterring potential users from this app, fearing any kind of logo complexity will have relevance to the complexity of the app itself.

Every successful app, or company for that matter has a slogan. Slogans are not as popular in the UK as in other places like the U.S.A or Europe, however, if I want my app to succeed on the global market, I will create a catchy slogan to compete with other app companies but also to entice potential users to the app. I decided to go with this slogan as it's short, to the point, and a clear description of the app's primary focus.

I designed the logo with my knowledge of the industry in mind using Adobe Photoshop to assist in the imagery, crafted the slogan based on the app's primary use on every stakeholder level, and determined the app name based on the name of the task itself.

This is the app logo in its final development stage on Photoshop.



This is the final version of the logo for my app '**Bottle up**'.
I believe the logo is consistent with the design ideology: simple and striking, recognizable and user-friendly.



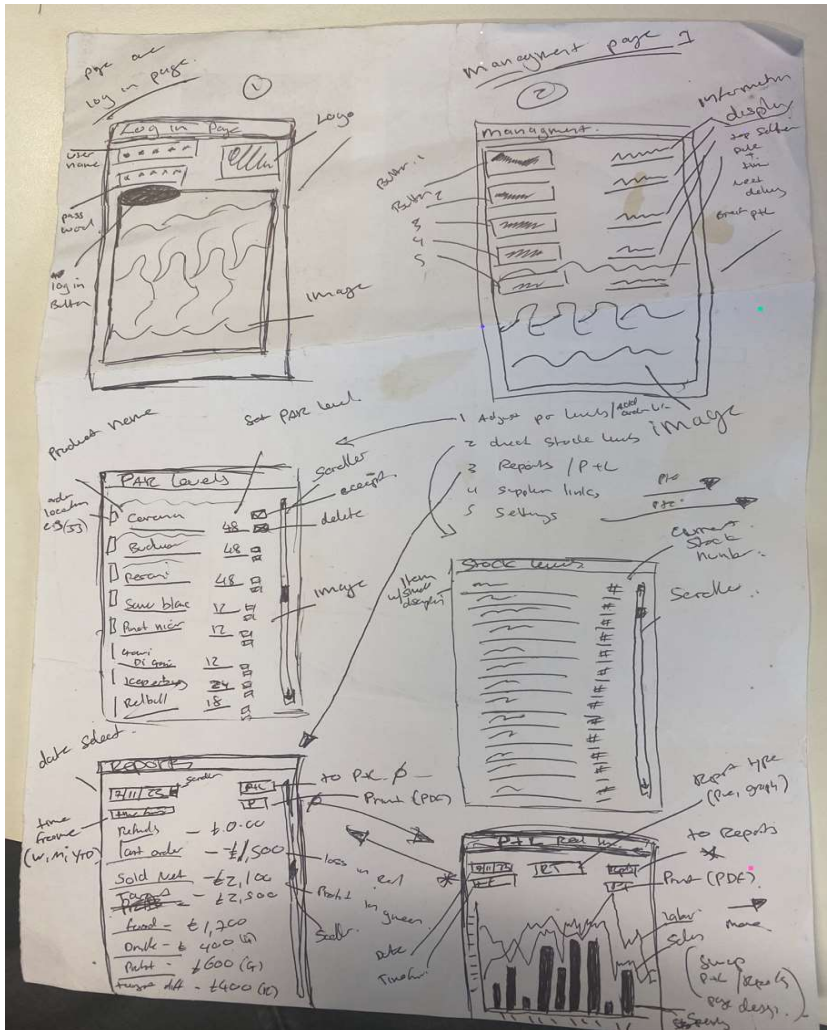
Bottle Up

FILLING THE FRIDGES SO YOU DON'T HAVE TOO

Design and Implimentaion

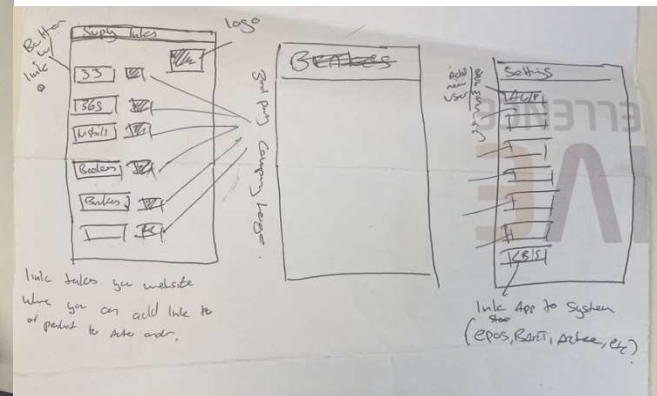
Now that the app conception, function, stakeholders, and logo have been set, it is time to begin designing the app pages and implementing them into a working prototype.

I began the app userface design using pen and paper, I find I have a lot more freedom of thought and ease of use using older methods compared to the modern way of design..



Even though the hand-drawn version is very rough and rudimentary, I find it to be the most engaging and helpful part of the design process. I believe this because it feels to me that there is no barrier between my mind and the creation, adding annotation and flow as I see fit, the only caps being my imagination and skill level.

Here I have designed and displayed multiple user screens including the log-in, the home page, par level adjustment, P&L graphs, selling trends, individual item stock levels, and automatic supplier orders.



For this assignment, I need only create three interface screens, which will be login, Homepage, and Par level adjustment, all screens will be displayed at a management stakeholder level.

Page 1: The Log in screen

The log-in screen, though simple in design thanks to the design ideology, has a hidden function. It will detect the user's stakeholder position based on login credentials.

For example, a management-level stakeholder will be directed to the full access and adjustment options of the app, showing displays such as the P&L, Suppliers, PAR levels, and more.

A staff-level stakeholder will only be referred to a limited version of the app, allowing them to input details such as stock waste or broken glass and crockery, but also showing real-time information on current stock levels and their position on the upselling ladder.



Page 2:

The Homepage screen

The homepage screen has closely followed the design ideology in the fact of its user-friendliness. All informative displays and adjustment options are displayed in short form, giving the user the ability to navigate the app and use its functions with relative ease.

The informative displays work in real-time against the company's chosen 'Point of sale' system, which can include but is not limited to EPOS, Aztec, BART, MPOS, Square, TouchBistro, etc.

These displays can be selected which will open a more detailed and thorough look at the information, ensuring the user gets accurate readings and reports.

These reports and information can be shared via email or in-app messenger, alternatively, the information can be printed for physical archiving.

Page 3:

The PAR level adjustment screen

The PAR level adjustment screen has been designed to allow the user to see which products, PAR levels, and suppliers are currently active quickly and easily on the 'Bottle up' app.

With simplicity and user-friendliness as the cornerstone of this development, the user will be able to adjust by selecting a supplier's bar and using a search tool or the alphabetized drop-down menu to efficiently scroll through to the product of choice, will be able to change the auto-order PAR level, whilst having the ability to select individual orders as well.

With the app being simple and user-friendly, under the hood the app will calculate the cost of all the selected items and cross the sum against internal systems and POS sales readings, giving the app accurate data for the P&L for the user to access in real-time.



CONCLUDING

This app is still in its very early stage of development, the app pages I have presented are working on MIT APP Inventor, however, at the current moment, only page navigation is possible. Once my coding skill improves or I find investment then the higher functions of this app will be operational. I believe this app will have a significant impact on the hospitality industry, possibly revolutionizing the way the industry connects suppliers with businesses, and then businesses to customers. I believe with further research and development the app's stakeholders could extend to the wider public, possibly creating a review system on local eateries, direct customer satisfaction results on individual supplier products, discounts, complimentary vouchers, etc. More analysis will be needed if the app finds this route to be beneficial. I have every intention of creating this app in the future and am looking into the requirements for patterning this idea.

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